UNIT – V

Vibrations, Turning Moment & Flywheels

Subject : Theory of Machines (23A03403)

Syllabus: Vibrations – introduction, DOF, types, free natural vibration (Newton & energy methods), damped (under, critical, over), forced vibration with and without damping, vibration isolation & transmissibility. Turning moment diagrams (steam engine, I.C. engine, multi-cylinder), crank effort, coefficients of fluctuation of energy & speed, flywheel design, flywheel for punching press.

1. Vibrations – Introduction

Vibration = repetitive (oscillatory) motion of a body about an equilibrium position. Causes : unbalance, elastic forces, external excitation. Effects : noise, fatigue, wear, discomfort, resonance failure ; useful in vibrators, musical instruments, ultrasonic cleaners.

- Terms : **period** T (time of one cycle), **frequency** $f = 1/T$ (Hz), **circular frequency** $\omega = 2\pi f$ (rad/s), **amplitude** X (maximum displacement), **phase** φ, **natural frequency** ω_n (frequency of free vibration), **resonance** (forcing frequency ≈ natural frequency).
- **Simple harmonic motion** : $x = X \sin(\omega t + \varphi)$; velocity $\dot{x} = X\omega \cos(\omega t + \varphi)$; acceleration $\ddot{x} = -\omega^2 x$ (proportional to displacement and directed to the mean position).

Degrees of freedom

Minimum number of independent coordinates required to define the motion : spring–mass (1), simple pendulum (1), double pendulum (2), two-rotor torsional system (2), a beam/continuous system (∞).

Types of vibration

Basis	Types	Remark
External force	Free (only initial disturbance) / Forced (periodic force)	ω_n / forcing ω

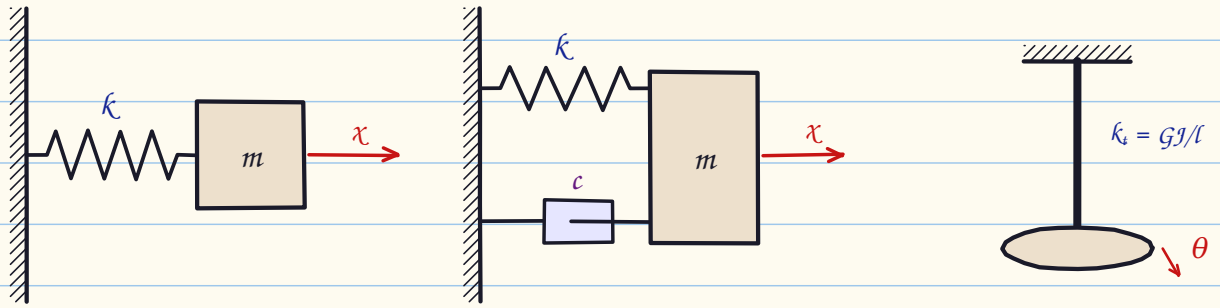
Basis	Types	Remark
Damping	Undamped / damped (viscous, Coulomb, structural)	energy dissipated
Motion	Longitudinal / transverse / torsional	spring-mass / beam / shaft
System	Linear / non-linear ; discrete / continuous	superposition holds for linear
Excitation	Deterministic / random	earthquake, road roughness

Equivalent stiffness

Case	Stiffness	Remark
Springs in parallel	$k = k_1 + k_2$	same deflection
Springs in series	$1/k = 1/k_1 + 1/k_2$	same force
Cantilever (end load)	$k = 3EI/l^3$	ω_n from m
Simply-supported beam, central load	$k = 48EI/l^3$	
Fixed–fixed beam, central load	$k = 192EI/l^3$	
Shaft (torsional)	$k_t = GJ/l = \pi G d^4/(32\,l)$	N·m/rad

2. Free Undamped Vibration (SDOF)

(a) Spring–mass (b) Spring–mass–damper (c) Torsional



$m\ddot{x} + kx = 0$ $m\ddot{x} + c\dot{x} + kx = F(t)$ $I\ddot{\theta} + k_t\theta = 0$

Newton's (equilibrium) method

Mass m on a spring of stiffness k, displacement x from the static equilibrium (weight is balanced by the static spring force, so it cancels) :

$$m\ddot{x} + kx = 0 \Rightarrow \omega_n = \sqrt{(k/m)}\ f_n = (1/2\pi)\sqrt{(k/m)}\ T = 2\pi\sqrt{(m/k)}$$

- Solution : $x = A\cos\omega_n t + B\sin\omega_n t = X\sin(\omega_n t + \varphi)$; $A = x_0$, $B = \dot{x}_0/\omega_n$; $X = \sqrt{(x_0^2 + (\dot{x}_0/\omega_n)^2)}$.
- **Static deflection** $\delta_{st} = mg/k \Rightarrow \omega_n = \sqrt{(g/\delta_{st})}$; $f_n = 0.4985/\sqrt{\delta_{st}}$ (δ_{st} in m).
- **Torsional** : $I\ddot{\theta} + k_t\theta = 0 \Rightarrow \omega_n = \sqrt{(k_t/I)}$; **simple pendulum** $\omega_n = \sqrt{(g/l)}$; **compound pendulum** $\omega_n = \sqrt{(g\,h/(k_G^2 + h^2))}$.
- **Two rotors on one shaft** (two-DOF) : node at distance $l_1 = l\,I_2/(I_1 + I_2)$ from rotor 1 ; $\omega_n = \sqrt{(\,k_t(I_1 + I_2)/(I_1\,I_2)\,)}$, $k_t = GJ/l$.

Energy (Rayleigh) method

Conservative system : total energy constant $\Rightarrow KE_{max} = PE_{max}$ (or $d(KE + PE)/dt = 0$).

$$\tfrac{1}{2}\,m\,\dot{x}_{max}^2 = \tfrac{1}{2}\,k\,x_{max}^2 \text{ with } \dot{x}_{max} = \omega_n x_{max} \Rightarrow \omega_n^2 = k/m$$

- **Effect of spring mass** m_s : effective mass $m + m_s/3 \Rightarrow \omega_n = \sqrt{(k/(m + m_s/3))}$. Pulley/rolling cylinder systems : include rotational KE.

Ex 1. $m = 5$ kg, $k = 2000$ N/m : $\omega_n = \sqrt{(2000/5)} = 20$ rad/s ; $f_n = 3.18$ Hz ; $T = 0.314$ s ; $\delta_{st} = 5\times9.81/2000 = 24.5$ mm. Check : $0.4985/\sqrt{0.0245} = 3.18$ Hz ☐.

Ex 2 (energy method). Uniform cylinder $m = 10$ kg, $R = 0.2$ m rolls without slipping, axle attached to spring $k = 3000$ N/m : $KE = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}(\tfrac{1}{2}mR^2)(\dot{x}/R)^2 = \tfrac{3}{4}\,m\,\dot{x}^2 \Rightarrow m_{eq} = 1.5\,m = 15$ kg $\Rightarrow \omega_n = \sqrt{(3000/15)} = 14.1$ rad/s.

Ex 3 (springs). Two springs 4000 and 6000 N/m : in parallel $k = 10\,000$ N/m ; in series $k = 2400$ N/m. A 10 kg mass on the series pair : $\omega_n = \sqrt{(2400/10)} = 15.5$ rad/s.

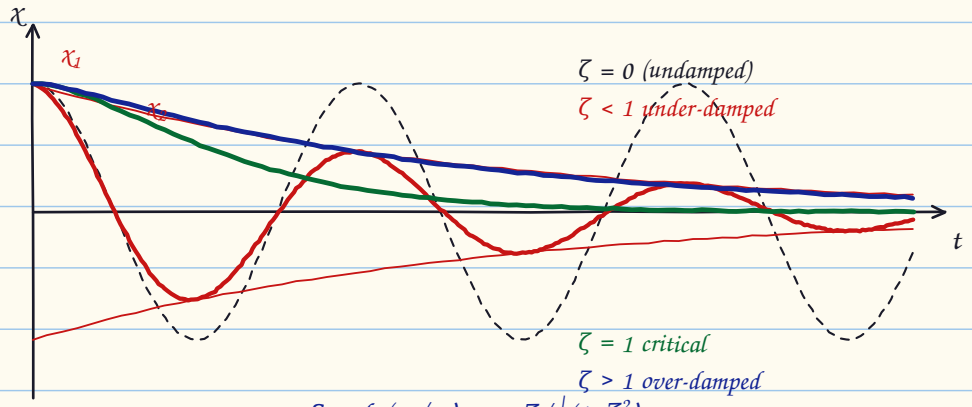
3. Free Damped Vibration

Viscous damping force $F_d = c\,\dot{x}$ (c in N·s/m) :

$$m\ddot{x} + c\dot{x} + kx = 0 \quad \zeta = c/c_c\ ,\ c_c = 2\sqrt{(km)} = 2m\omega_n \text{ (critical damping)}$$

Characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \Rightarrow s = -\zeta\omega_n \pm \omega_n\sqrt{(\zeta^2 - 1)}$.

Free vibration with viscous damping



$\delta = \ln(x_1/x_2) = 2\pi\zeta/\sqrt{(1-\zeta^2)}$

Case	Solution	Motion
Under-damped $\zeta < 1$	$x = X\, e^{-\zeta\omega_n t}\} \sin(\omega_d t + \varphi)$, $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$	oscillatory, decaying amplitude
Critically damped $\zeta = 1$	$x = (A + Bt)\, e^{-\omega_n t}\}$	no oscillation, fastest return to rest
Over-damped $\zeta > 1$	$x = A\, e^{s_1 t}\} + B\, e^{s_2 t}\}$ ($s_{1,2}$ real negative)	aperiodic, sluggish return

- Under-damped : successive peaks decrease in geometric progression. **Logarithmic decrement** (ratio of two successive amplitudes, period $T_d = 2\pi/\omega_d$) :

$$\delta = \ln(x_1/x_2) = \zeta\omega_n T_d = 2\pi\zeta/\sqrt{(1-\zeta^2)} \approx 2\pi\zeta \text{ (small } \zeta) \quad \delta = (1/n)\ln(x_0/x_n) \text{ for } n \text{ cycles}$$

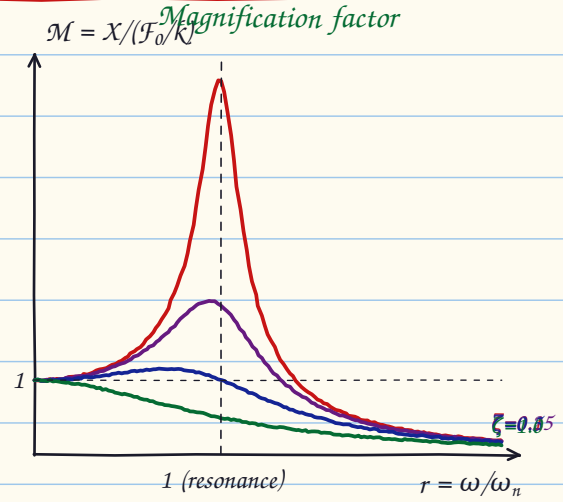
- Use δ to measure ζ experimentally : $\zeta = \delta/\sqrt{(4\pi^2 + \delta^2)}$.
- Damped frequency $\omega_d < \omega_n$ (very little difference for $\zeta < 0.2$). Time constant $\tau = 1/(\zeta\omega_n)$.
- **Coulomb (dry-friction) damping** : amplitude decreases linearly, by $4\mu N/k$ per cycle ; motion stops when $kx \le \mu N$. Structural (hysteretic) damping $\propto$ displacement.
- Practical use of critical damping : door closers, galvanometers, gun recoil, suspension ($\zeta \approx 0.2$–0.4 in cars), instrument pointers.

Ex 4. $m = 10$ kg, $k = 4000$ N/m, $c = 40$ N·s/m : $\omega_n = 20$ rad/s ; $c_c = 2\sqrt{(4000\times10)} = 400$; $\zeta = 0.1$; $\omega_d = 20\sqrt{0.99} = 19.9$ rad/s ; $\delta = 2\pi(0.1)/\sqrt{0.99} = 0.631 \Rightarrow x_1/x_2 = e^{0.631} = 1.88$. Amplitude after 5 cycles $= x_0/1.88^5 = 4.3$ % of x_0.

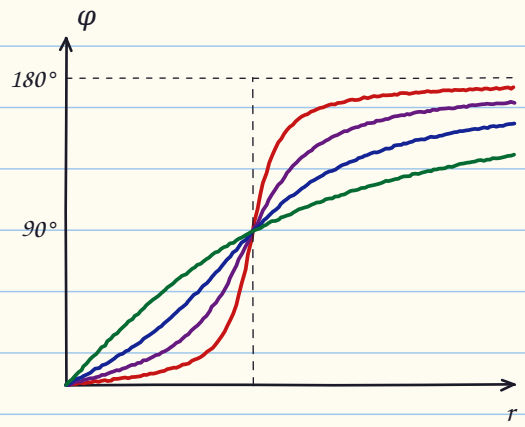
Ex 5. Observed amplitude ratio over 10 cycles is $2:1 \Rightarrow \delta = (1/10)\ln 2 = 0.0693$ $\Rightarrow \zeta = 0.0693/\sqrt{(39.5 + 0.0048)} = 0.011$.

4. Forced Vibration

Magnification factor



Phase angle



(a) Without damping

Harmonic force $F_0 \sin\omega t$: $m\ddot{x} + kx = F_0 \sin\omega t$. Steady-state amplitude :

$$X = (F_0/k) / (1 - r^2)\ r = \omega/\omega_n$$

- For $r < 1$ the motion is in phase with the force ; for $r > 1$ it is 180° out of phase (sign negative). At $r = 1$: resonance – amplitude theoretically infinite (X grows linearly with time).

(b) With viscous damping

$m\ddot{x} + c\dot{x} + kx = F_0 \sin\omega t \Rightarrow$ steady-state $x = X \sin(\omega t - \varphi)$:

$$X = (F_0/k) / \sqrt{(\ (1 - r^2)^2 + (2\zeta r)^2\ )}\ \tan\varphi = 2\zeta r / (1 - r^2)$$

- **Magnification factor** $M = X/(F_0/k) = 1/\sqrt{((1 - r^2)^2 + (2\zeta r)^2)}$. $M \to 1$ for $r \to 0$ (static) ; $M \to 0$ for $r \gg 1$ (mass cannot follow). At resonance ($r = 1$) : $M = 1/(2\zeta)$ and $\varphi = 90°$ irrespective of damping. Peak occurs at $r = \sqrt{(1 - 2\zeta^2)}$: $M_{max} = 1/(2\zeta\sqrt{(1 - \zeta^2)})$.
- Damping matters only near resonance ; far from resonance the system is mass or stiffness controlled. The total response = transient (dies out) + steady-state. Q-factor $Q = 1/(2\zeta)$; half-power bandwidth $\Delta\omega = 2\zeta\omega_n$.

Ex 6. $m = 20$ kg, $k = 8000$ N/m, $c = 80$ N·s/m, $F_0 = 100$ N, $\omega = 15$ rad/s : $\omega_n = 20$; $r = 0.75$; $c_c = 800 \Rightarrow \zeta = 0.1$. $X = (100/8000)/\sqrt{((1 - 0.5625)^2 + (0.15)^2)} = 0.0125/0.4625 =$ 27.0 mm ; $\varphi = \tan^{-1}(0.15/0.4375) =$ 18.9°. At resonance $X = 0.0125/0.2 = 62.5$ mm.

(c) Rotating unbalance (e.g. unbalanced motor)

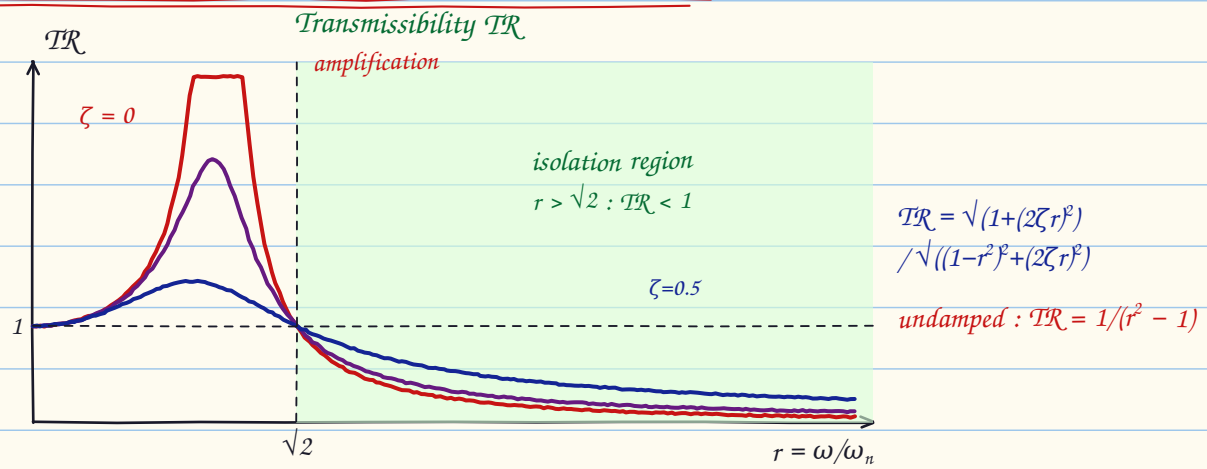
Total mass M, unbalanced mass m_u at eccentricity e, speed ω : forcing amplitude $F_0 = m_u\, e\, \omega^2$.

$$X = (m_u\, e/M)\, r^2 / \sqrt{(1 - r^2)^2 + (2\zeta r)^2}\ \ X \to m_u e/M \ (r \gg 1)$$

(d) Support (base) excitation & whirling of shafts

- Base moves $y = Y \sin\omega t$: absolute response ratio $X/Y = \sqrt{(1 + (2\zeta r)^2)} / \sqrt{((1 - r^2)^2 + (2\zeta r)^2)}$ = transmissibility. Vehicle suspensions, seismographs ($r \gg 1$), accelerometers ($r \ll 1$).
- **Critical speed of a shaft** (whirling) $N_c = 30/\pi \times \sqrt{(g/\delta_{st})}$ = natural frequency of transverse vibration ; operate well above or below it.

5. Vibration Isolation & Transmissibility



Isolation reduces the force transmitted from a machine to the foundation (force isolation) or the motion transmitted from the foundation to the equipment (motion isolation).

- Force transmitted $F_T = \sqrt{((kX)^2 + (c\omega X)^2)} \Rightarrow$ **transmissibility ratio** :

$$TR = F_T/F_0 = \sqrt{(1 + (2\zeta r)^2)} / \sqrt{(1 - r^2)^2 + (2\zeta r)^2}\ \ (\text{undamped} : TR = 1/(r^2 - 1))$$

- $TR < 1$ only for $r > \sqrt{2}$. So the natural frequency of the mount must be much lower than the disturbing frequency : $f_n < f/\sqrt{2}$. Isolation efficiency $\eta = (1 - TR) \times 100$ %.
- **Effect of damping** : increases isolation near resonance ($r \approx 1$) but reduces it for $r > \sqrt{2}$ (TR becomes larger) – a compromise : $\zeta \approx 0.05$–0.2 for passing through

resonance during start-up.

- Methods : steel springs, rubber pads, cork, felt, pneumatic mounts, inertia blocks (increase m), dynamic vibration absorber (tuned mass damper) to cancel a given forcing frequency.

Ex 7. Machine 100 kg, 1200 rpm (ω = 125.7 rad/s) ; required TR = 0.25 (efficiency 75 %), undamped isolators : $r^2 = 1 + 1/TR = 5 \Rightarrow r = 2.236 \Rightarrow \omega_n$ = 56.2 rad/s $\Rightarrow k = m\omega_n^2$ = 316 kN/m ; static deflection $\delta_{st} = g/\omega_n^2$ = 3.1 mm ; transmitted force = 0.25 F_0.

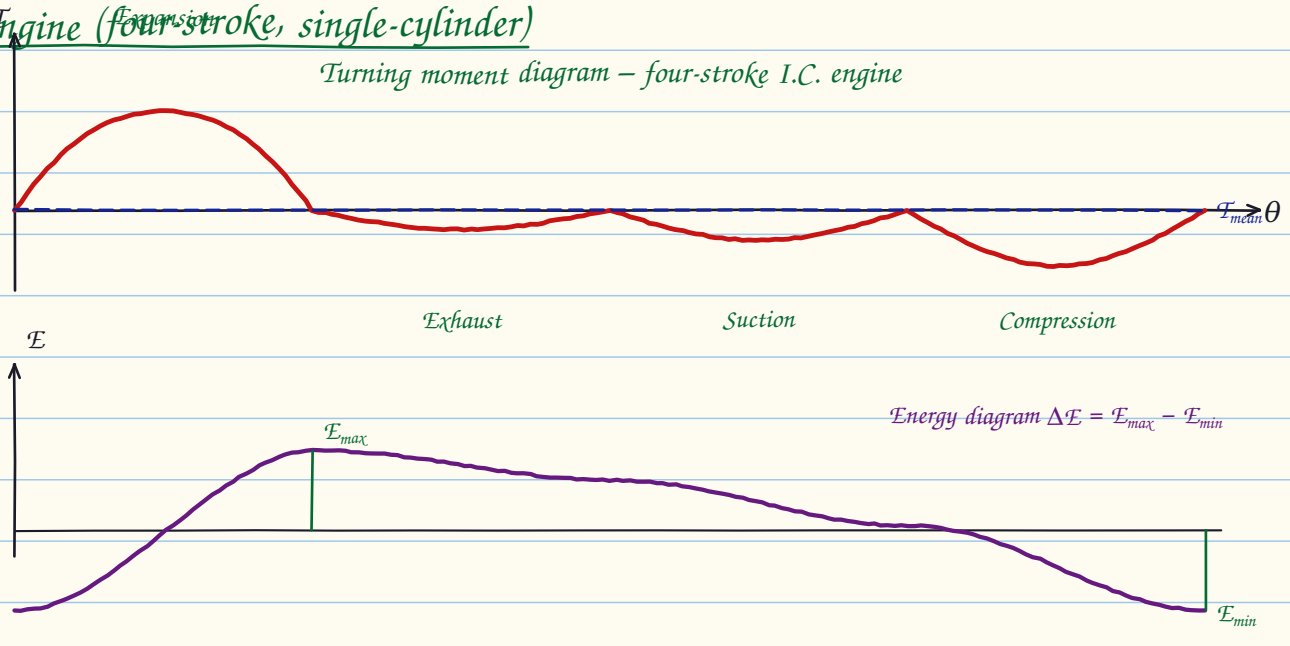
Ex 8. Amplitude of transmitted force with ζ = 0.1 at r = 3 : $TR = \sqrt{(1 + 0.36)}/\sqrt{(64 + 0.36)} = 1.166/8.02 = 0.145$ (vs 0.125 undamped) – damping slightly worse above $\sqrt{2}$.

6. Turning Moment Diagrams

Turning moment (crank effort) $T = F_p\, r \sin(\theta + \varphi)/\cos\varphi$ varies with crank angle because the piston force varies. The turning-moment diagram plots T against θ for one cycle.

- **Mean (average) torque** T_{mean} = (work done per cycle)/(angle of cycle) : 2π for steam engine / two-stroke, 4π for four-stroke ; $P = T_{mean}\,\omega$. The area under the curve over the cycle = work per cycle.

I.C. Engine (four-stroke, single-cylinder)



- Cycle = 720° : expansion (positive large torque), exhaust, suction, compression (negative torque – work done on the gas). Only one stroke in four does positive

work ⇒ very large fluctuations ⇒ heavy flywheel.

- **Steam engine (double-acting)** : torque produced in both strokes ; the diagram has two humps per revolution (θ = 0–180° and 180°–360°), never negative ⇒ small fluctuation.
- **Multi-cylinder engines** : individual curves, shifted by the phase angle (720°/number of cylinders for four-stroke), are added ⇒ smoother resultant, smaller flywheel (fluctuation ratio falls from ≈ 2 for single-cylinder four-stroke to ≈ 0.2 for four-cylinder).
- **Crank effort** = torque on the crankshaft = T (above) ; reaction torque on frame equal and opposite.

Fluctuation of energy & speed

- Above T_{mean} the engine produces more energy than the load needs ⇒ flywheel speeds up ; below T_{mean} it gives energy back ⇒ slows down.

$\Delta E = E_{max} - E_{min}$ (maximum fluctuation of energy = difference between the highest and lowest points of the cumulative-energy diagram)

- **Coefficient of fluctuation of energy** $C_E = \Delta E$ / (work done per cycle). **Coefficient of fluctuation of speed** $C_s = (\omega_1 - \omega_2)/\omega_{mean} = (N_1 - N_2)/N_{mean}$, $\omega_{mean} = (\omega_1 + \omega_2)/2$. Coefficient of steadiness $m = 1/C_s$.

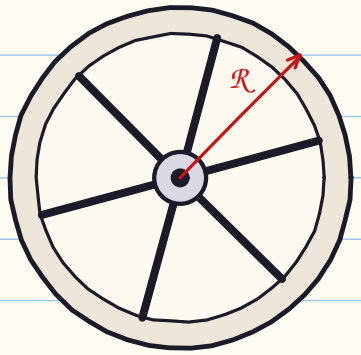
Machine	C_s
Crushing machines	0.2
Punching, shearing	0.03 – 0.05
Pumps, machine tools	0.03 – 0.05
Weaving / spinning	0.01 – 0.02
Automobile engines	0.02 – 0.05
Generators, turbines	0.002 – 0.01
Alternators / dc generators for lighting	0.0035 – 0.005

7. Flywheel

A heavy rotating wheel (rim + arms + hub) mounted on the crankshaft that stores kinetic energy when the engine produces excess energy and releases it when the engine produces less, keeping the speed within limits. (Differs from a governor : flywheel controls the **cyclic** speed variation within one cycle ; governor controls the **mean** speed

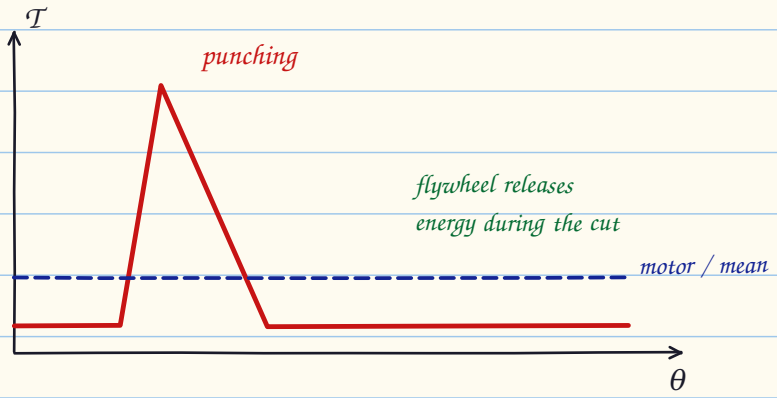
over many cycles with load changes.)

Flywheel (rim, arms, hub)



$\Delta E = I\,\omega^2\,C_s = m\,k^2\,\omega^2\,C_s$

Punching-press torque demand



$E_{punch} - E_{motor\ in\ punching\ time} = \Delta E$

$$\Delta E = I\,(\omega_1^2 - \omega_2^2)/2 = I\,\omega^2\,C_s = m\,k^2\,\omega^2\,C_s \approx m\,v^2\,C_s \;(\text{rim},\ k \approx R,\ v = \omega R)$$

Design of the flywheel

- (1) Find ΔE from the turning-moment (energy) diagram ; (2) choose C_s ; (3) $I = \Delta E/(\omega^2 C_s)$; (4) assume mean rim radius R (or v) and find rim mass $m = I/R^2$ (neglect arms and hub, ≈ 90 % of I in the rim) ; (5) rim cross-section $A = b\,t = m/(\rho\,2\pi R)$ ($b = (1–2)\,t$) ; (6) check hoop stress.
- **Hoop (centrifugal) stress** in the rim :

$$\sigma = \rho\,v^2 \Rightarrow v_{max} = \sqrt{(\sigma_{allow}/\rho)}$$

Cast iron $\rho \approx 7200\ kg/m^3$, $\sigma_{allow} \approx 6\ MPa \Rightarrow v \approx 29\ m/s$; cast steel $\rho = 7800$, $\sigma \approx 20–40\ MPa \Rightarrow v \approx 50–70\ m/s$.

- Arms (4–8 elliptical spokes) carry the torque (bending stress $\sigma_b = T/(Z)$) ; hub keyed to shaft ; large flywheels are split / segmental. Flywheels of high-speed energy storage use composite rims.

Ex 9. Energy areas of one cycle (J) about T_{mean} : +52, −124, +92, −140, +85, +35. Cumulative : E_0+52 → E_0−72 → E_0+20 → E_0−120 → E_0−35 → E_0. $E_{max} = E_0$+52 ; $E_{min} = E_0$−120 ⇒ ΔE = 172 J. N = 600 rpm (ω = 62.83), C_s = 0.02 : $I = 172/(62.83^2 \times 0.02)$ = 2.18 kg·m² ; for rim R = 0.3 m : m = 2.18/0.09 = 24.2 kg.

Ex 10. Rim, cast iron : R = 0.6 m, N = 300 rpm ⇒ v = 18.85 m/s ⇒ $\sigma = 7250 \times 18.85^2$ = 2.58 MPa (safe, below 6 MPa).

Ex 11. 4-stroke engine, 20 kW at 300 rpm : $T_{mean} = 20000/31.42 = 637$ N·m ; work per cycle = $T_{mean} \times 4\pi$ = 8000 J (= 20 kW × 0.4 s). If ΔE = 1.0 × work per cycle (illustrative) and $C_s = 0.02$: $I = 8000/(31.42^2 \times 0.02) = 405$ kg·m² ; maximum / minimum speeds $N_{1,2} = 300(1 \pm 0.01) = 303 / 297$ rpm.

8. Flywheel for Punching Press

Punching, shearing, riveting presses need a large energy during a very short part of the cycle but little in the rest ⇒ the motor is sized for the average power and the flywheel supplies the peak.

- Work done per punch (assuming force falls linearly from F_{max} to 0 during the plate thickness t) : $E_{punch} = \tfrac{1}{2} F_{max}\, t$, $F_{max} = \pi\, d\, t\, \tau_u$ (circular hole dia d in plate of thickness t, ultimate shear strength τ_u).
- If the cycle time is t_c (time per punch) and the punching (actual cutting) time is t_p : energy supplied by the motor during the punch = $(E_{punch}/t_c) \times t_p$; **energy to be supplied by the flywheel** :

$$\Delta E = E_{punch} - E_{punch}\, t_p/t_c = E_{punch}\,(1 - t_p/t_c)$$

- Then $I = \Delta E/(\omega^2 C_s)$; for efficiency η_m the motor power = $E_{punch}/(t_c\, \eta_m)$. The maximum speed fall during the punch : $\omega_2 = \omega(1 - C_s/2)$. The flywheel regains its speed during the remainder of the cycle.

Ex 12. d = 20 mm hole in 15 mm plate, τ_u = 350 MPa : $F_{max} = \pi(0.02)(0.015)(350\times10^6) = 330$ kN ; $E_{punch} = \tfrac{1}{2} \times 330\,000 \times 0.015 = 2474$ J. 30 holes/min (t_c = 2 s), punching time t_p = 0.4 s ⇒ motor supplies 2474 × 0.4/2 = 495 J ⇒ $\Delta E = 1979$ J. N = 300 rpm (31.42 rad/s), $C_s = 0.02$: $I = 1979/(31.42^2 \times 0.02) = 100$ kg·m² ; mean motor power = 2474/2 = 1237 W (≈ 1.24 kW).

- Concept : a lighter flywheel needs a bigger motor ; heavier flywheel – smaller motor but more starting power and hoop stress.

9. Additional Solved Problems

P1 (natural frequency of a beam). Cantilever steel (E = 200 GPa) 0.5 m long, $I = 8\times10^{-9}$ m⁴, end mass 5 kg : $k = 3EI/l^3 = 3(200\times10^9)(8\times10^{-9})/0.125 = 38.4$ kN/m ⇒ $\omega_n = \sqrt{(38400/5)} = 87.6$ rad/s ⇒ $f = 13.9$ Hz.

P2 (torsional). Disc $I = 0.5\ kg{\cdot}m^2$ on steel shaft $d = 20\ mm$, $l = 0.8\ m$, $G = 80\ GPa$: $J = \pi d^4/32 = 1.571\times10^{-8}\ m^4$; $k_t = GJ/l = 1571\ N{\cdot}m/rad \Rightarrow \omega_n = \sqrt{(1571/0.5)} =$ 56.1 rad/s.

P3 (damping ratio). A system has $c = 1.2\ kN{\cdot}s/m$, $k = 100\ kN/m$, $m = 50\ kg$: $c_c = 2\sqrt{(100000\times50)} = 4472 \Rightarrow \zeta = 0.268$ (under-damped). $\omega_n = 44.7\ rad/s$; $\omega_d = 43.1\ rad/s$.

P4 (resonance). A motor of mass 120 kg on springs of total $k = 120\ kN/m$: $\omega_n = 31.6\ rad/s = 302\ rpm$; to avoid resonance operate at 1440 rpm ($r = 4.77$) $\Rightarrow$ TR (undamped) $= 1/(4.77^2 - 1) =$ 0.046 (95 % isolation).

P5 (flywheel speed). $C_s = 0.03$ about $N = 150\ rpm \Rightarrow N_{max} = 150(1 + 0.015) = 152.25$; $N_{min} = 147.75\ rpm$.

Short answers – Quick Revision

- **What is resonance?** Forcing frequency = natural frequency $\Rightarrow$ amplitude limited only by damping ($X = F_0/(2\zeta k)$).
- **Why is isolation poor for $r < \sqrt{2}$?** Transmissibility ≥ 1 – the mount amplifies the force.
- **Why does a single-cylinder engine need a heavy flywheel?** Only the expansion stroke produces positive torque $\Rightarrow$ big ΔE.
- **Difference between flywheel and governor?** Flywheel – cyclic fluctuation (energy reservoir) ; governor – mean speed vs load (control of fuel/steam).
- **What does $\zeta = 1$ mean?** Fastest return to equilibrium without overshoot – critically damped.

10. Quick Formula Sheet – Unit V

Item	Formula
Natural frequency	$\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta_{st})}$; torsional $\sqrt{(k_t/I)}$
Damping ratio	$\zeta = c/(2\sqrt{(km)})$; $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$
Log decrement	$\delta = 2\pi\zeta/\sqrt{(1-\zeta^2)} \approx 2\pi\zeta$
Forced amplitude	$X = (F_0/k)/\sqrt{((1-r^2)^2+(2\zeta r)^2)}$
Transmissibility	$TR = \sqrt{(1+(2\zeta r)^2)}/\sqrt{((1-r^2)^2+(2\zeta r)^2)}$; isolation if $r > \sqrt{2}$
Mean torque	$T_{mean} = W_{cycle}/(2\pi$ or $4\pi)$

Item	Formula
Flywheel	$\Delta E = I\omega^2 C_s$; $C_s = (\omega_1 - \omega_2)/\omega_{mean}$
Rim stress	$\sigma = \rho v^2$
Punching	$E = \frac{1}{2} F\, t$; $\Delta E = E(1 - t_p/t_c)$

11. Important Questions

- Define DOF and classify vibrations. Derive the natural frequency of a spring–mass system by Newton's and energy methods.
- Derive the equation of motion of a damped SDOF system and discuss under-damped, critically damped and over-damped cases; define logarithmic decrement.
- Derive the steady-state response of a damped forced system ; draw magnification factor & phase curves ; define resonance.
- Define transmissibility ; derive the expression and show that isolation needs $r > \sqrt{2}$; describe isolation methods.
- Draw and explain the turning-moment diagrams of a steam engine, four-stroke I.C. engine and multi-cylinder engine.
- Define coefficient of fluctuation of energy and speed ; derive $\Delta E = I\omega^2 C_s$; explain the design of a flywheel and the flywheel for a punching press.